

Open-Set RF Fingerprint Verification: Separability vs the Operating-Point Gap

Separability is real; the hard-authentication operating point is not, a measured boundary study

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Abstract

RF fingerprinting identifies wireless devices by the microscopic hardware imperfections that manufacturing tolerance leaves on the analog transmit chain, even across units of the same model. Framed as an open-set verification problem, distinguishing an enrolled genuine device from an impostor that never appeared in training or calibration, separability is real. Across three public datasets (WiSig, INRIA PLA, WIDEFT) we measured single-packet ($K=1$) AUROC in the 0.70-0.87 range, clearly above chance (0.5). But once the question shifts to hard authentication, the true accept rate at the low false-accept operating point an access-control gate must actually enforce, that signal collapses. On INRIA (CC BY 4.0, commercially usable) TAR at FAR=0.1% was only 2.9%; on WiSig's cross-receiver split it fell to 0.5%. The fact that the same gap reproduces across two datasets under different licenses (WiSig is non-commercial CC BY-NC-SA, INRIA is CC BY) suggests this is a property of the task, not an artifact of one dataset. Ten-packet fusion ($K=10$) narrows the gap substantially but does not close it (TAR is still 61.6% at FAR=0.1%), and self-supervised pretraining raised AUROC modestly under low-label conditions while, in our measurement, the strictest operating point's TAR moved in the opposite direction. The honest conclusion is that today's public data and public methods do not support standing RF fingerprinting up as a hard access-control gate. What can be sold with integrity is not a pass/fail gate but a continuous risk score fused with other signals.

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1 Abstract

RF fingerprinting identifies wireless devices by the microscopic hardware imperfections that manufacturing tolerance leaves on the analog transmit chain, even across units of the same model. Framed as an open-set verification problem, distinguishing an enrolled genuine device from an impostor that never appeared in training or calibration, separability is real. Across three public datasets (WiSig, INRIA PLA, WIDEFT) we measured single-packet ($K=1$) AUROC in the 0.70-0.87 range, clearly above chance (0.5). But once the question shifts to hard authentication, the true accept rate at the low false-accept operating point an access-control gate must actually enforce, that signal collapses. On INRIA (CC BY 4.0, commercially usable) TAR at FAR=0.1% was only 2.9%; on WiSig’s cross-receiver split it fell to 0.5%. The fact that the same gap reproduces across two datasets under different licenses (WiSig is non-commercial CC BY-NC-SA, INRIA is CC BY) suggests this is a property of the task, not an artifact of one dataset. Ten-packet fusion ($K=10$) narrows the gap substantially but does not close it (TAR is still 61.6% at FAR=0.1%), and self-supervised pretraining raised AUROC modestly under low-label conditions while, in our measurement, the strictest operating point’s TAR moved in the opposite direction. The honest conclusion is that today’s public data and public methods do not support standing RF fingerprinting up as a hard access-control gate. What can be sold with integrity is not a pass/fail gate but a continuous risk score fused with other signals.

2 Background

Wireless devices built to the same design and the same bill of materials do not behave identically. Manufacturing tolerance in the analog front end, oscillator phase noise, ADC/DAC nonlinearity, IQ imbalance, power amplifier response curves, differs slightly from unit to unit, and that difference leaves a reproducible trace on the transmitted waveform. RF fingerprinting (often shortened to RFFI) extracts that trace as a feature and asks which physical device sent this signal.

The appeal is obvious. No extra credential or certificate has to be provisioned; the identity signal comes straight from the physical layer. Credentials can be stolen or cloned; the intuition is that hardware manufacturing tolerance is much harder to replicate. That intuition drives proposals ranging from detecting unregistered devices on a factory wireless network, to authenticating industrial IoT gateways, to gating access on medical-device networks. The problem is that nearly all of these proposals are backed by closed-set classification accuracy, how accurately the system can tell which known transmitter, among a fixed list, sent a given signal. Access control asks a different question: when a device that never appears on the enrolled list tries to impersonate an enrolled identity, can the system reject that attempt at a low false-accept rate? High closed-set accuracy guarantees nothing about that question. This paper measures that gap directly and honestly.

We reopen this question because many headline numbers in the literature are the product of post-hoc selection: fusing many packets, sweeping several scoring methods, and picking the best combination after seeing the results. An AUROC of 0.999 obtained that way measures something different from the deployment condition that actually matters, a single decision made the instant one packet arrives, against a threshold fixed in advance. This paper strips that selection out and reports only what survives under a pre-registered single-packet ($K=1$) condition and a pre-registered multi-packet ($K=10$) condition.

3 Evaluation Methodology

Open-set verification has two stages. A subset of known devices is used for enrollment, training, and threshold calibration; the remaining devices are held out entirely as unknown. At test time, the system must correctly reject signals from unknown devices while correctly accepting signals from enrolled devices it has legitimately seen. The critical discipline is that unknown-device data must never touch calibration or training. Any leakage inflates performance optimistically. Every experiment in this study calibrated thresholds only on a held-out validation split of enrolled devices, exposing unknown-device data solely at test time.

Three metrics matter here. Equal Error Rate (EER) is the single number where the false-accept and false-reject rates coincide, a compact summary of overall separability. AUROC integrates discriminative power across all possible thresholds, describing separability before any threshold has been chosen. Neither metric directly answers what an access-control deployment needs to know. A real deployment fixes one false-accept-rate target (FAR) and asks what fraction of genuine devices are accepted at that operating point (TAR, true accept rate). AUROC can be high while TAR@FAR is low, and that gap is this paper’s central finding. We report AUROC and EER alongside TAR at FAR=10%, 1%, and 0.1% throughout. FAR=0.1% sits close to what a hard access-control gate typically demands, and the TAR at that point is effectively “what fraction of legitimate users get locked out if this system is turned into a gate.”

Point estimates from small device counts (8 to 24 known devices) carry real sampling variability. We report device-level bootstrap confidence intervals to make that explicit. The INRIA K=1 AUROC point estimate of 0.868, for example, carries a 95% CI of [0.810, 0.932], wide because the underlying sample is only 8 physical devices, not a computation error. This paper’s negative conclusion holds even at the optimistic end of that interval.

The number of fused packets (K) is also a methodological choice with real consequences. K=1 corresponds to the actual access-control scenario where a single packet arrives and a decision cannot wait: no latency budget for accumulation. Raising K to 10 or 50 averages scores across multiple packets and substantially improves separability, but it costs time to accumulate those packets and makes the decision boundary more predictable to an adversary. We report K=1 as the primary result and a pre-registered K=10 as supplementary context. Post-hoc, best-of-many K=50 numbers reported elsewhere in the literature are cited only as external context, never as this paper’s own finding.

This post-hoc-selection concern is not abstract. A separate leakage audit of a related public result found that its headline K=50 fusion number had been the best of 18 combinations (six K values times three scorers) selected after seeing performance on the held-out split, not a combination declared in advance and then measured once. A number obtained that way is unlikely to reproduce in deployment, where results cannot be seen before a scoring method is chosen. That is why this paper’s headline results are limited to K=1 and a single pre-registered K=10.

4 Data and Setup

This study uses three public datasets. We deliberately did not narrow to one, so we could check directly whether the same result reproduces across datasets under different licenses and capture conditions.

Dataset	License	Device population	Capture conditions
WiSig (Hanna, Karunaratne, Cabric, 2022)	CC BY-NC-SA 4.0 (non-commercial)	20 known devices, real over-the-air capture	Multiple receivers, multiple days, cross-receiver split
INRIA PLA (Zenodo record 18268648)	CC BY 4.0 (commercially usable)	8 known + 4 held-out devices, 12 COTS IoT transmitters	Single BladeRF receiver, single session, 2.4 GHz ISM
WIDEFT (Zenodo record 4116383)	CC BY 4.0 (commercially usable)	24 known consumer device models + 14 held-out	Single USRP B210 receiver, real WiFi capture

WiSig is the most widely used benchmark in this field, but its CC BY-NC-SA license restricts it to non-commercial research validation. Any commercially defensible conclusion requires re-running the same protocol on data that permits commercial use. We therefore independently reproduced the same open-set verification protocol on INRIA PLA (CC BY 4.0). This re-measurement was not merely a license workaround; it was a direct test of whether the gap was a quirk of WiSig’s specific noise profile or device population. As the results below show, the gap did not close: INRIA actually posted a higher K=1 AUROC than WiSig (0.868 vs. 0.729), yet TAR@FAR=0.1% remained under 3%.

WIDEFT is a WiFi RF fingerprint dataset of consumer electronics, phones, laptops, routers, streaming boxes. It is a reasonable proxy for the task of enrolling authorized devices and rejecting unregistered ones on a shared PHY-layer population, but it proves nothing about the actual RF emission characteristics of medical devices or about regulatory compliance. Here we deliberately used a different method family: 28 classical DSP features (IQ imbalance, CFO, instantaneous frequency/phase/amplitude statistics, higher-order cumulants, spectral bands) with a LightGBM/Mahalanobis classifier rather than a learned embedding, to check whether the same failure pattern appears even without deep learning.

The learned pipeline’s backbone is an ECAPA-TDNN, an architecture widely used for speaker verification, adapted to raw IQ, trained with an ArcFace-style angular-margin loss combined with momentum contrastive learning (MoCo). At verification time, distance between the query embedding and an enrolled gallery of embeddings served as the score for computing AUROC, EER, and TAR@FAR. Separately, we tested whether self-supervised pretraining helps under low-label conditions: pretraining on masked-IQ-span reconstruction, then fine-tuning on the same verification task, compared against random initialization across three seeds.

5 Results

The table below reports measured results under the pre-registered conditions. All thresholds were calibrated only on the enrolled-device validation split; unknown-device data was exposed only at test time.

Condition	K	AUROC	EER	TAR@FAR=10%	TAR@FAR=1%	TAR@FAR=0.1%
WiSig, cross-receiver, random init (3-seed mean)	1	0.729	0.334	-	12.1%	4.2%
WiSig, cross-receiver, masked-IQ SSL pretrain (3-seed mean)	1	0.752	0.314	-	10.9%	2.6%
WiSig, cross-receiver, random init, pre-registered fusion	10	0.953	0.116	-	33.9%*	14.5%*
INRIA PLA, same receiver, same session	1	0.868	0.215	63.0%	20.8%	2.9%
INRIA PLA, same receiver, same session, pre-registered fusion	10	0.997	0.024	99.95%	91.4%	61.6%
WiSig, cross-receiver, separate domain-adaptation run	1	0.695	0.344	-	-	0.5%

Condition	K	AUROC	EER	TAR@FAR=10%	TAR@FAR=1%	TAR@FAR=0.1%
WIDEFT consumer- device proxy, classical DSP + distance- based	1	0.519	0.480	-	0.7%	0.1%

* WiSig K=10 row uses one representative seed; unlike INRIA we did not run a separate 3-seed aggregate for the pre-registered K=10 fusion arm.

Three patterns stand out. First, K=1 AUROC ranges from 0.52 (WIDEFT, near chance) to 0.87 (INRIA) across three datasets and two method families (learned embeddings and classical DSP), yet TAR@FAR=0.1% never exceeded 5% in any condition. Even INRIA, the highest-AUROC condition measured, posted a TAR of 2.9% at that point. High AUROC and usable access control are separate claims.

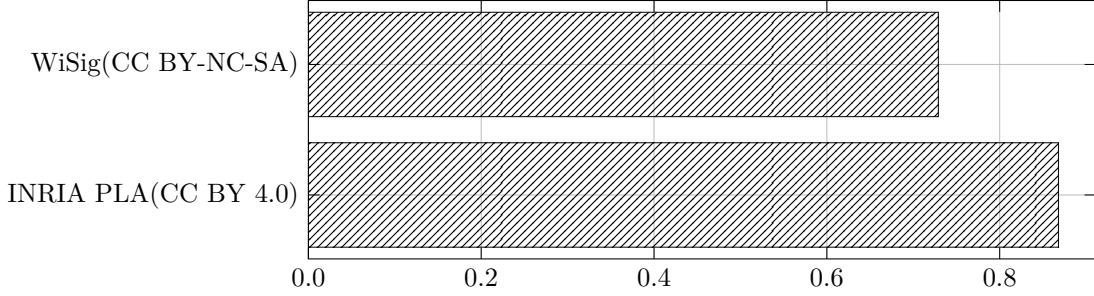


Figure 1: Separability vs the operating-point gap

Second, multi-packet fusion (K=10) narrows this gap substantially but does not close it. On INRIA, K=10 pushes TAR to 91.4% at FAR=1%, close to an access-control threshold, but tightening FAR to 0.1% drops TAR back to 61.6%. Even after accumulating ten packets, the strictest false-accept target still locks out more than one in three legitimate users. The nonlinearity of this collapse as FAR tightens is stark.

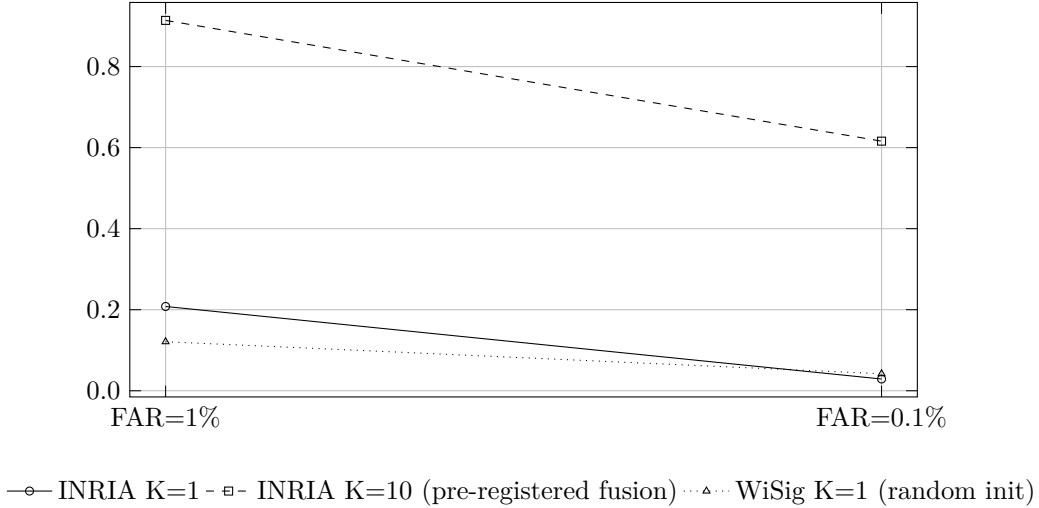


Figure 2: TAR collapses as FAR tightens

Third, self-supervised pretraining’s effect diverged by metric. Masked-IQ reconstruction pretraining raised K=1 AUROC from 0.729 to 0.752, a gain of +0.0235 (3-seed mean, consistent in direction across all three seeds).

But under the same condition $\text{TAR@FAR}=1\%$ fell from 12.1% to 10.9%, and $\text{TAR@FAR}=0.1\%$ fell from 4.2% to 2.6%. All three seeds moved in the same direction (SSL lower) at both FAR points, which argues against this being random noise. Because AUROC integrates discriminative power across the entire threshold range, an improvement in the relaxed-threshold region can mask a degradation in the strict-threshold region. This result independently reconfirms, through a concrete intervention, this paper’s central caution against equating a rising AUROC with improved authentication performance.

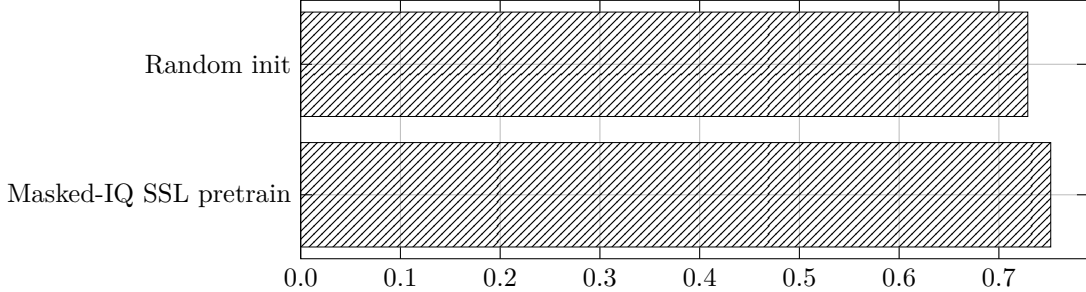


Figure 3: SSL pretraining: AUROC rises while TAR at the strict operating point falls

Fourth, two standard mitigations we tried also failed to close the gap. Gradient-reversal-layer (GRL) domain adaptation performed slightly worse turned on than off (0.727 vs. 0.737 AUROC), and source-free contrastive adaptation lowered $K=1$ AUROC by 0.030. This suggests the gap is not a minor loss-function or scoring artifact but a bottleneck in the feature space itself.

6 Implications

The most direct implication is unambiguous. Today’s public data and public methods, learned metric embeddings, classical DSP with distance-based classifiers, self-supervised pretraining, domain adaptation, do not support standing up a single-decision hard authentication gate ($K=1$, FAR target at or below 1%). This conclusion is not tied to one dataset or one backbone. The same failure pattern reproduced across a non-commercially-licensed dataset (WiSig) and a commercially usable one (INRIA), across same-receiver and cross-receiver conditions, and across a learned pipeline and a classical DSP pipeline.

Multi-packet fusion is a useful mitigation, not a solution. Pre-registered $K=10$ fusion approaches a practical threshold at a relaxed false-accept target ($\text{FAR}=1\%$), but it requires accumulating packets over time, which is incompatible with instant-decision scenarios, and it still falls short at the strictest target ($\text{FAR}=0.1\%$). The near-1.0 AUROC numbers sometimes cited in the literature for $K=50$, best-of-many-scorers combinations are not what this paper reproduced; they are reported to have been selected after seeing results without pre-registration, and should be treated as measuring a different, less deployable condition.

The honest product form implied by this evidence is not a hard gate but a risk-score layer. RF fingerprint verification should output a continuous confidence value, fed into a fusion platform alongside other signals, network-layer identity, behavioral baselines, geofencing, where the final decision tolerates a moderate false-accept rate because another signal or a human adjudicates borderline cases. That is a less dramatic claim than a pass/fail gate, but it is the only form this measured evidence honestly supports today.

We also state the conditions under which this conclusion could change. A narrower, more homogeneous enrolled population (same vendor, same procurement batch) might reduce cross-vendor heterogeneity and improve separability. Re-measurement on a different PHY and environment, such as an actual IoMT capture, could produce different numbers. What we measured is what today’s public data and public methods show; we do not claim this permanently forecloses every RF fingerprinting application. We do claim that any proposal to sell hard authentication today should be required to present evidence at the level of rigor shown here.

Building a risk-score product on this evidence implies several concrete design decisions. First, the consumer of the confidence value should be a fusion platform, not the end user directly; gating access on the RF score alone exposes the exact false-accept and false-reject rates measured here to real users. Second, thresholds need periodic

recalibration as the receiver, environment, or enrolled population changes, and that recalibration must reuse the enrolled validation split, not live unknown-device traffic, or the leakage this paper warns against reappears in production. Third, this score should function as one layer of defense in depth rather than a standalone gate: a low-confidence attempt should trigger additional verification, a step-up challenge or an operator alert, rather than an outright block, which is the design that matches the current level of evidence.

Finally, we are explicit about what this study did not measure: cross-day and cross-environment drift, robustness under active spoofing attack, and performance at enrolled populations much larger than tested here (50+ devices). A larger enrolled population likely increases the chance of an unknown device colliding with some enrolled identity at any fixed threshold, so the numbers reported here are, if anything, more optimistic than a real large-scale deployment would see.

7 Data and Reproducibility

All three datasets used in this study are public. WiSig (Hanna, Karunaratne, Cabric, 2022) is CC BY-NC-SA 4.0, restricted to non-commercial research validation. INRIA PLA (Zenodo record 18268648) and WIDEFT (Zenodo record 4116383) are both CC BY 4.0 and commercially usable. The evaluation protocol fixed the following in advance: thresholds are calibrated only on the enrolled-device validation split, with unknown-device data exposed only at test time; the known/unknown split is device-disjoint; AUROC and EER are always reported alongside TAR at FAR=10%/1%/0.1%; K is reported at K=1 as the primary result with a single pre-registered K as supplementary context, never re-selected after seeing results; and small device-count point estimates carry device-level bootstrap confidence intervals. Evaluation scripts and exact hyperparameters are available on request.

8 References

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